Windows Connectivity Troubleshooting Runbook

Technician decision guide | Companion to SOP-IT-014

Document ID	**RB-IT-014A**
Revision	1.0 — sample / not approved
Effective date	Not applicable — educational demonstration
Audience	Tier 1 IT support, shift lead, endpoint/network administrator
System scope	Windows 11 managed endpoints — fictional Bayhaven Services
Owner / approver	Fictional IT Operations / no real approvals

PORTFOLIO DEMONSTRATION — Synthetic tickets, endpoints, contacts and outcomes. This document is not an authorized enterprise runbook.

1. When to use and safety gates

Use this runbook after creating the ticket and identifying the user-impact scope. Perform only diagnostics permitted by organizational policy. It prioritizes low-risk checks; reset commands are not default steps. See SOP-IT-014 for approvals, privacy and escalation requirements.

2. Symptoms to branch mapping

Reported symptom	First differentiating observation	Likely workstream
No network icon / cable disconnected	Physical cable, adapter and Windows Network & internet status	Physical/adapter
Connected but no websites	Two destinations; service status; IP/gateway; DNS	Client/DNS/upstream
VPN fails but public websites work	VPN error, clock, client and login flow	VPN/IAM
One app fails	Service status, error, account scope	Application support
Drops repeatedly	Timestamp, cable/SSID, repeatability, second device	Network performance

3. Command reference — safe interpretation

Command	What it checks	What it does not prove
ipconfig /all	Local IP, DHCP, gateway and DNS configuration	That DHCP/DNS services are healthy
ping <gateway>	ICMP response from gateway if permitted	A timeout automatically means gateway down
nslookup example.com	Direct DNS query using configured DNS server	That browser/Windows resolution path is identical
Resolve-DnsName example.com	Windows DNS resolution behavior	That all application network paths are healthy
ipconfig /flushdns	Clears DNS resolver cache; modifies state	Should only run with authorization and recorded reason

4. Worked fictional incident: DNS mismatch

Ticket INC-IT-0264: a user can access an internal service by approved IP address, but the corresponding DNS name does not resolve. Device WIN11-042 is a synthetic endpoint. The examples below illustrate evidence-writing and are not actual command outputs.

1. 08:14 — User reports the incident; agent verifies device and confirms only one employee is affected. Ticket status: Investigating.

2. 08:19 — Windows reports Ethernet connected. ipconfig /all lists an IPv4 address, gateway and an unexpected DNS server for this fictional environment. Agent redacts the full configuration before placing it in a general ticket.
3. 08:23 — Gateway ping replies; agent notes that ICMP result supports reachability, not end-to-end connectivity.
4. 08:26 — nslookup for the fictional service fails using the configured server. A second device on the same network resolves the service correctly. Agent documents the discrepancy without changing DNS.
5. 08:30 — Agent routes INC-IT-0264 to Network Services with the expected/observed DNS server evidence, impact and verification steps; status remains Open, awaiting resolver review.

5. Ticket escalation example

Field	Synthetic example
Summary	Windows 11: internal hostname fails to resolve; Ethernet connected
Impact / urgency	Single user; critical internal app unavailable; priority per approved matrix
Evidence	ipconfig /all config summary; nslookup failure; known-good comparison
Steps performed	Status, physical check, read-only command tests; no configuration changes
Current status	Escalated / open — resolution not claimed
Owner / next action	Network Services to verify assigned DNS policy and respond to Tier 1
Customer update	Explained observed issue and next contact without promising an ETA

6. User communication — example

“Thanks for walking through the checks with me. Your computer shows an Ethernet connection, but the service name is not resolving as expected. I have documented what we observed and sent the results to our network team for further investigation. I will update this ticket when they respond; I have not changed your network settings.”

7. Recheck and closure

- After resolver action, repeat the specific failing user workflow and capture results.
- Record user confirmation and whether service remains stable; avoid asserting a root cause not evidenced.
- If DNS configuration changes recur, open a problem-investigation recommendation rather than prematurely closing the underlying issue.

8. Reference and validation limits

Commands and cautious interpretation cross-checked against Microsoft Support/Microsoft Learn references listed in SOP-IT-014. The fictional timeline is illustrative; no live Windows lab test was executed for this case.

REGULATORY / TECHNICAL REVIEW ADDENDUM — November 2026

Independent technical writing portfolio enhancement. Additional validation gates and examples are illustrative, not authorized production instructions. Regulated applicability varies by product, process, record and organization.

Read-only command examples with interpretation

Command: ipconfig /all. Use: identify active interface, IPv4 addressing, default gateway, DHCP and DNS server. Caution: do not upload raw unredacted configuration publicly; avoid assuming inactive or virtual adapters are the active path.

Command: nslookup example.com. Use: query DNS using the configured or specified resolver; a successful response does not prove the application, VPN or proxy is functioning. Command: Resolve-DnsName example.com in PowerShell; differences can reflect differing resolver paths or settings.

Command: ping <approved-gateway>. Use: reachability clue only. Timeout cannot prove failure because ICMP may be filtered; record network and service context. Use Test-NetConnection only if available and approved, selecting a permitted host and service port.

Scenario matrix and safe next action

No link detected -> verify cable, link LEDs and approved adapter; do not open network cabinets. Self-assigned address -> investigate DHCP path or adapter and escalate. DNS name only fails -> compare approved internal resolver, VPN and managed DNS settings; do not arbitrarily set public DNS. Several users impacted -> route major incident; avoid repeated local resets.

A successful DNS query with failed application login suggests service/authentication diagnostics rather than generic cable failure. A successful IP ping but failed website access can reflect blocked ports, proxy, TLS or service health and requires additional evidence.

Complete sample incident evidence package

INC-IT-0264 (synthetic): user reported 08:14; read-only interface capture 08:19; sanctioned DNS test 08:26; escalated 08:30. Record expected DNS profile, observed resolver class, impact scope, command identifiers and redaction note. Outcome: AWAITING NETWORK TEAM; no verified restoration yet.

A later fictional completion record would need change approval, corrected configuration evidence, successful service-name lookup, actual application access and user confirmation. Do not label resolved solely because a change was attempted.

Document-specific reference basis and status

Microsoft Ethernet connection troubleshooting (full URL in portfolio review report)

Microsoft nslookup reference (full URL in portfolio review report)

NIST SP 800-61 Rev. 3 (Apr 2025) (full URL in portfolio review report)

NIST SP 800-53 Rev. 5 security and privacy controls (full URL in portfolio review report)

ISO/IEC/IEEE 26514:2022 information for software users (full URL in portfolio review report)

W3C WCAG 2.2 color use guidance (full URL in portfolio review report)

Status: technical/editorial review against publicly available references completed at a portfolio-example level; no licensed-standard full-text conformance audit, live environment testing, site QA approval or legal certification performed.

Appendix: November 2026 standards and documentation applicability review

PUBLIC SYNTHETIC PORTFOLIO SAMPLE — This appendix records source-checking and editorial scope boundaries. It does not make the fictional organization regulated, certify a system, or replace actual company policies and competent quality/legal review.

Windows troubleshooting and cybersecurity response references (2026 review)

- Microsoft Learn product documentation governs Windows command meanings and Teams product settings; confirm software version, tenant policy, permissions and endpoint environment before issuing operational steps.
- NIST SP 800-61 Rev. 3 (2025) provides cyber incident response guidance; a routine DNS or DHCP service interruption is not automatically a security incident.
- NIST CSF 2.0 and SP 800-53 control families help frame asset management, access, logging, change control and response; they are not universal private-company regulations.
- After a suspicious Teams support call, do not assume screen sharing equals remote control, or remote control equals malware persistence; investigate authorized remote session evidence and endpoint alerts.
- Preserve audit trails, use read-only diagnostics first, require change authorization before resets and protect credentials/PII in help-desk records.
- Editorial QA: command outputs are illustrative, test procedures are proposed, and the portfolio does not represent a live verified Windows deployment.

Reference links and recheck notes

https://csrc.nist.gov/pubs/sp/800/61/r3/final

https://www.nist.gov/cyberframework

https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/ipconfig

https://learn.microsoft.com/en-us/microsoftteams/meeting-who-present-request-control

Verified against published authoritative source pages as of November 2, 2026. No proprietary standards were audited clause-by-clause. Actual application must be assessed based on product, system use, region, and record type. Proposed procedures have not been executed in a production environment.

Independent second-pass document review | November 2, 2026

REVIEWED PORTFOLIO DEMONSTRATION — this appendix records a document-specific editorial and source-scope check. The main procedure and forms are fictional and not approved for production, executed testing or regulated use. Original source versions are retained as editable files in this portfolio.

A. Document-specific findings and checks

Review objective	2026 reviewer correction / expectation	Verification evidence
Command interpretation	ipconfig, ping, nslookup, Resolve-DnsName each tests a defined layer and has limitations.	Do not infer network reachability solely from blocked ICMP or single successful DNS query.
Change authorization	DNS flush, adapter reset, network reset and remote assistance are approval-gated and reversible where feasible.	List evidence before/after and a rollback step.
Security red flags	Unexpected remote tool, external Teams IT impersonation, suspicious download -> security queue.	No instruction to bypass endpoint security or install unapproved tools.

B. Applicable regulatory / guidance boundaries

Review objective	2026 reviewer correction / expectation	Verification evidence
Primary scope	Windows network and service-desk diagnostics.	Non-security connectivity incidents are not automatically cyber incidents; record symptoms and escalation criteria.
Security reference	NIST 800-61r3 April 2025; NIST CSF 2.0; SP 800-53 release 5.2.0	Voluntary unless independently adopted/mandated; use defensible evidence capture and owner-approved changes.
Safe actions	Read-only checks first; approval before flush/reset, driver changes, remote-control or account resets.	A DNS test alone cannot prove all application or internet paths work; confirm user-observed service restoration.

C. Evidence and sign-off boundaries

Status: Editorial second-pass reviewed; technical procedure not executed; company-specific SOP approval not obtained.

All tests, screenshots, users, suppliers, dates, IDs, incident observations, performance metrics and signatures in the demonstration are synthetic.

Next practitioner action: confirm source applicability for the actual jurisdiction and product; conduct SME/user testing and accessibility review; update controlled revision and approval fields.

No color, watermark, revision number, PDF export or reference link by itself demonstrates regulatory compliance.

D. Authoritative primary-source references

NIST 800-61r3: https://csrc.nist.gov/pubs/sp/800/61/r3/final

NIST 800-53 release 5.2.0: https://csrc.nist.gov/News/2025/nist-releases-revision-to-sp-800-53-controls

US Access Board Section 508: https://www.access-board.gov/ict/about/